

ER Diagram

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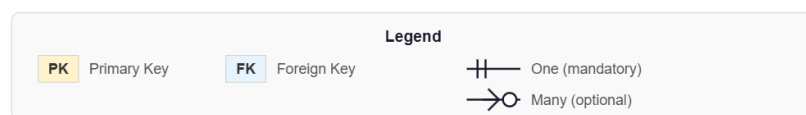
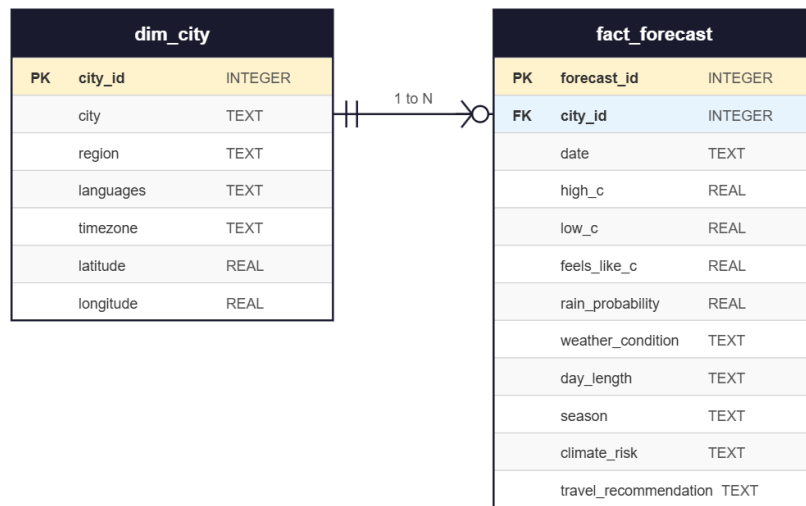
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Course: 692 Pipeline to Insights

Project: World Capitals Climate Dashboard

Database: SQLite (via SQLAlchemy)

Entity Relationship Diagram



Relationship Summary

The diagram above shows two entities connected by a one to many relationship. The **dim_city** table is the parent entity. The **fact_forecast** table is the child entity. One city in **dim_city** can have many forecast records in **fact_forecast**, linked through the **city_id** foreign key.

Key Constraints

- **city_id** in **dim_city** is the Primary Key (PK). It uniquely identifies each city.
- **city_id** in **fact_forecast** is the Foreign Key (FK). It references **dim_city.city_id** to link each forecast to its city.

- forecast_id in fact_forecast is the Primary Key (PK). It uniquely identifies each forecast record.
- A unique constraint on city_id and date prevents duplicate forecast records for the same city on the same day.

Cardinality

One row in dim_city corresponds to exactly 7 rows in fact_forecast, one for each day in the 7 day forecast window. With 10 cities the total is 10 rows in dim_city and 70 rows in fact_forecast.